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Sex differences in the relationship of cognition and brain in midlife adults at risk for dementia

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Introduction: Two-thirds of Alzheimer's Disease (AD) cases occur in women¹. Compared to men, women exhibit more rapid cognitive decline and brain atrophy in the presence of AD-related neuropathology². It is now acknowledged that AD processes are present decades before the onset of clinical symptoms³. However, whether there are sex differences in cognition-brain structure coupling and how AD risk affects their relationships in midlife remain unclear. In this study, we investigated associations between sex, AD risk, brain structure and cognition.

Methods: Participants: 701 cognitively unimpaired middle-aged participants (40–59 years) were recruited in the PREVENT-Dementia study from five study sites: West London, Edinburgh, Cambridge, Oxford and Dublin. In total, 614 participants (233 M/ 381 F) who had completed cognitive, clinical and structural Magnetic Resonance Imaging (sMRI) data were included in this study. Risk factors: The Apolipoprotein E [APOE] ϵ 4 risk is determined by ≥ 1 APOE ϵ 4 allele. The Cardiovascular Risk Factors Aging and Dementia (CAIDE) score is calculated based on eight variables⁴, with higher scores indicating greater risk. MRI data acquisition and processing: sMRI data were acquired using a T1-weighted MPRAGE sequence (TR = 2.3 s, TE = 2.98 ms, 160 slices, flip angle = 9°, voxel size = 1 mm³ isotropic). Freesurfer v7.1.0 was used for data processing⁵. The recon-all pipeline was run with default settings for each participant. The cortical thickness in 68 regions was quantified based on the Desikan-Killiany atlas⁶. The global cortical thickness (CT) was obtained by averaging the values from the bilateral hemispheres for each participant. We chose nine regions of interest (ROIs) relating to AD from previous studies^{7–9}, and the mean CT within each ROI was calculated by averaging the values from the bilateral hemispheres. Statistical analyses: Linear regression models were used to investigate the association of CAIDE with global CT and episodic and relational memory separately. To investigate the relationships between CT and cognition, and the moderating role of the sex variable, we used the linear regression model for episodic and relational memory, with global CT and nine regional CT as the independent variable (in independent models), sex as the moderator, and age, years of education, ICV and study sites included as covariates. The same moderation regression analysis was repeated in APOE ϵ 4+ and APOE ϵ 4– groups separately to further delineate the sex moderation effect on cognition-CT relationships in participants belonging to different risk groups. Multiple comparisons correction was carried out using the Bonferroni method.

Results: CAIDE was negatively associated with global CT (Fig. 1a), and negatively associated with episodic and relational memory (Fig. 1b). We didn't find a significant association between global CT and cognition (Fig. 2a) but found a sex-specific coupling of global CT and episodic and relational memory (Fig. 2b). Males showed a positive association between global CT and cognition, while females showed no relationship between them. Such sex-specific coupling between global CT and cognition was absent in APOE ϵ 4 carriers, and only shown in APOE ϵ 4 non-carriers (Fig. 2c, 2d). Among the nine ROIs, we only found a significant sex-specific coupling of episodic and relational memory and precuneus CT after Bonferroni correction. Furthermore, cognition was decoupled from the precuneus CT in APOE ϵ 4 carriers, and only coupled with CT of precuneus in APOE ϵ 4 non-carriers (Fig. 2e, 2f).

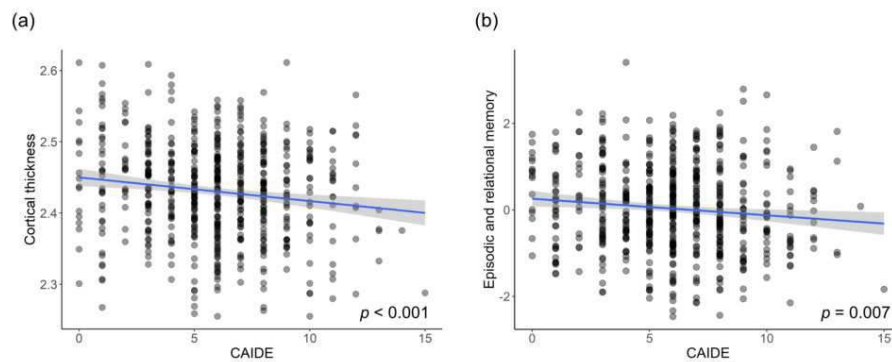


Figure 1. Association of CAIDE score with (a) cortical thickness, and (b) episodic and relational memory performance. On the x axis, higher scores represent higher CAIDE score, and on the y axis, higher scores represent thicker cortical thickness and better episodic and relational memory performance. Scatter plots show unadjusted values. Full regression statistic for cortical thickness, after controlling for head size and study sites: β (SE) = -0.004 (0.001), $p < 0.001$. Full regression statistic for episodic and relational memory: β (SE) = -0.04 (0.01), $p = 0.007$. Abbreviation: CAIDE, Cardiovascular Risk Factors, Aging and Dementia.

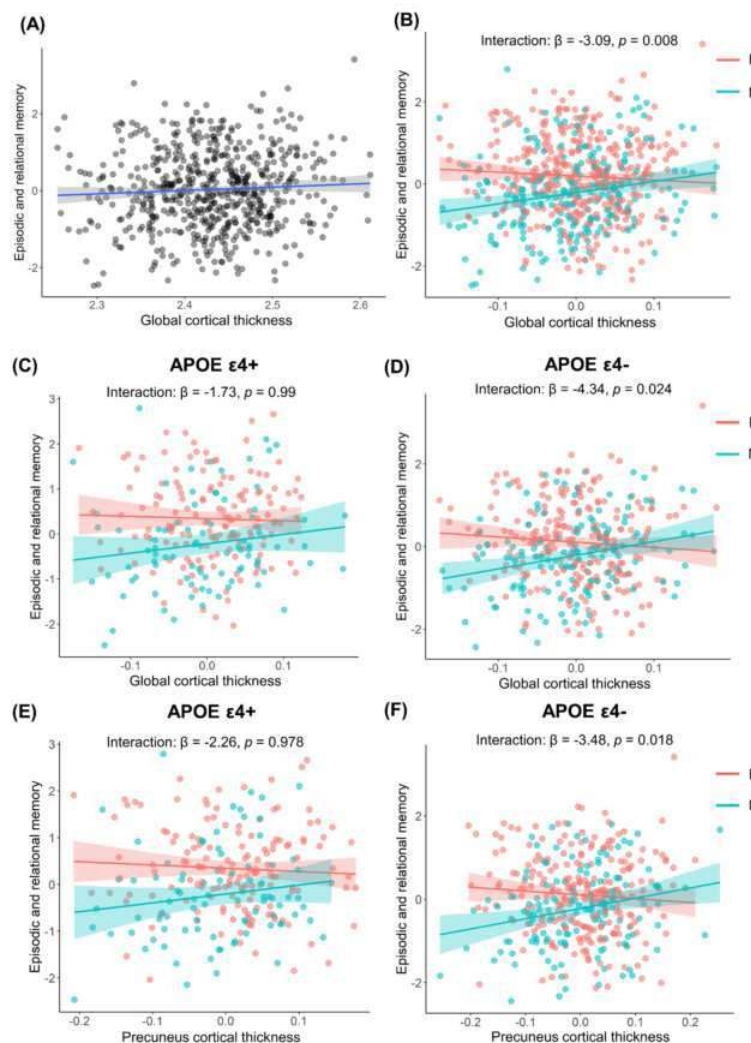


Figure 2. Associations between risk, cortical thickness and cognition in midlife. (a) No significant association between global cortical thickness and cognition. Interactions of sex and global cortical thickness on cognition in (b) the full cohort, (c) the APOE $\epsilon 4+$ group, and (d) the APOE $\epsilon 4-$ group. Interactions of sex and precuneus cortical thickness on cognition in (e) the APOE $\epsilon 4+$ group, and (f) the APOE $\epsilon 4-$ group. The scatter plot shows unadjusted values, but the statistical significance was based on the regression models where we controlled for covariates. P values for the interaction effects were Bonferroni corrected for multiple comparisons.

Conclusions: We found inherent sex-specific differences in the coupling between brain structure and cognition. Our results suggest that these sex-specific differences are being eroded by APOE $\epsilon 4$ carriership in mid-life. Longitudinal follow-up in this